

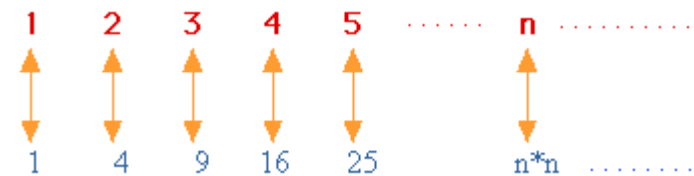
Historical notes on cardinality

Introduction

Historical Notes on Cardinality

In the 9th century the star worshipper and mathematician Thabit ibn Qurra, born in 826 in Harran, Mesopotamia (now Turkey), argued that **infinity could be split into two, three, or any number of parts, each of which would then have the same size as the whole**: there is for example, an infinite number of evens, but also of odds, so each half of infinity is infinity, and so on.

In 1638 Galileo argued that “equal”, “greater” and “less” can’t apply to infinite quantities because a line segment contains an infinity of points, so a longer segment should contain more than that infinity, which is impossible. And again, each natural number has its square matched with it:



Therefore there must be exactly as many square numbers as there are naturals. But at the sametime they should be fewer since *the squares grow ever sparser as you go along...*

Putting the question aside for a long time seemed to be the best way of coping with it.

The first one who revolutionized our understanding of the infinite was the German mathematician Georg Cantor (1845-1918). He introduced several basic notions: *set* and *cardinality of a set*.

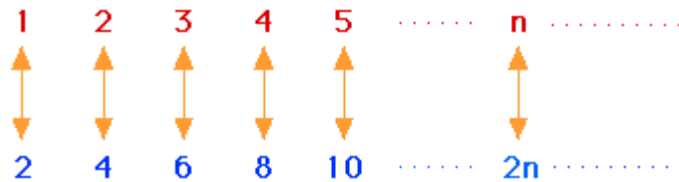
The cardinality of the set {rabbit, fox, wolf } is 3, and so is the cardinality of the set [-25, 30 180] because there is 1-1 correspondence between the elements of these sets:

Rabbit « -25

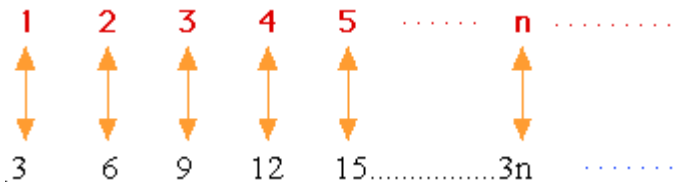
Fox 30

Wolf « 180

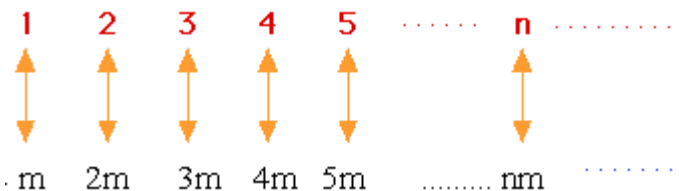
What about infinite sets? We could establish 1-1 correspondence between the natural and the even numbers, viz. each natural is perfectly matched with its double:



To say that these matchings-up show that there are as many of one kind as of the other needs, of course, a very bold leap of thought. We are taking each sort as a completed whole: **all the naturals** match up perfectly with **all the evens** (or with all the odds). Stop short and the correspondence breaks down e.g. there are only 50 evens among the first 100 naturals. The multiples of 3 are even *thinner* on the ground than those of 2 and still, there are as many of them as of the naturals (they are selected from):



More generally: for any natural number m , there are just as many multiples of m as there are natural numbers:



If you ask: *how many is that?* we could say: *the set N of the natural numbers and the set mN of the multiples of m have the same cardinality, as established by 1-1 correspondence we made.* Since we count by means of the natural numbers, we could also say that both sets are **countable**.

The crucial ideas in mathematic are always so simple as to seem intuitively clear: **sets and making 1-1 correspondences between their members**. You needn't even know how to count to do this. In the 4th century a nomadic army from the East rode through the Caucasus into Armenia. "No one could number the vastness of the cavalry contingents" wrote a contemporary chronicler. "So every man was ordered to carry a stone so as to throw it down... so as to form a mound...And wherever they passed, they left such markers at every crossroad along their way." During some battles the soldiers would set down a stone and those who survived would take each his stone back home.

An 1-1 correspondence between the members of the sets might be hard to find, but once revealed, the two sets between which it ran had to have the same cardinality. Conversely, if somehow you proved that no 1-1 correspondence could exist between the members of two sets, then their cardinalities would have to be different.

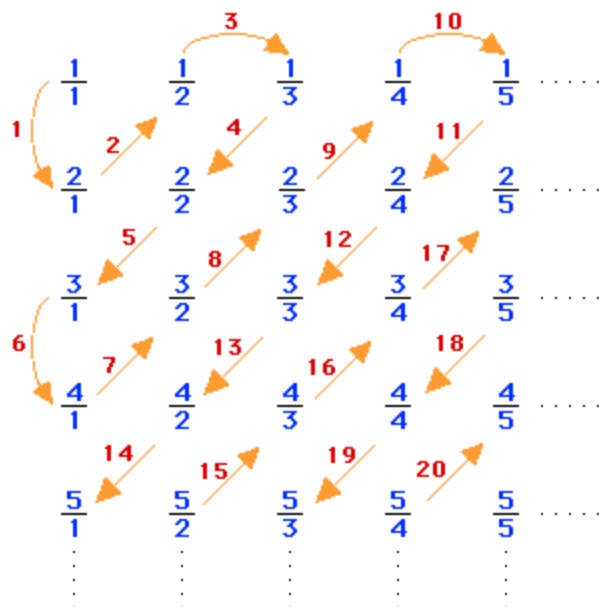
Having found that any infinity sequence of the naturals is countable (not a great surprise as it first seemed, if you think about it, since such a sequence will have a first term, then a second, third and so on – and ordering them thus in effect counts them), we could look in the opposite direction – at a set that includes the natural numbers, viz. the set $\mathbb{Z}$ of integers (naturals, 0 and negatives whole numbers). Is it possible to put this set too in 1-1 correspondence with its subset $\mathbb{N}$?

Yes, but with a little bit of imagination, a possible match up would be the following one:



This shows that the integers have the same cardinality as the naturals.

What about the rational numbers – isn't it obvious that they should be more than the integers and thus – forming a *more powerful* infinity?.. After all between any two fractions will lie another whereas you could always tell the neighbors of any integer. In other words, any set as numerous as the naturals is countable, but how could you possibly count the positive rationals? It looks as if you can't even start – between 0 and any candidate you name, another will appear... Clearly an order from smaller to larger is not possible because of the ceaseless in-betweens. But Cantor looked at the fractions from another angle:



Let us start from the upper-left corner. To continue horizontally or vertically would leave “vast quantity” of fractions untouched. So Cantor decided to “travel all over them in zigzag along diagonals” mode. Following the arrows you could put into 1-1 correspondence all the positive rationals with the natural numbers. Of course some fractions will appear more than

once, but we can easily get rid of the duplicates. Thus a possible 1-1 correspondence is the following one:



This strange sequence of rational numbers makes sense only when you see its 2D source, and what it shows is that the set of positive rationals contains no larger infinity than the infinity of the natural numbers.

Or, we could say that **“how many” has nothing to do with “how dense”!**

What did Cantor achieve? He matched up the members of the set N of the natural numbers with the members of the set Z of the integers, and then – with the members of the set of the rational numbers. Thanks to the notions “set” and “1-1 correspondence” the notion of “infinity” had not many vague meanings but one. The next candidate for a set with a bigger cardinality than the natural numbers was the set R of the real numbers. Cantor managed to show that there was no 1-1 correspondence between the sets N and R by a proof of contradiction. Namely, he assumed that the real numbers in the interval $(0,1)$ are countable, i.e. they could be arranged in a sequence as follows:

$0.a_{11} a_{12} a_{13} \dots$

$0.a_{21} a_{22} a_{23} \dots$

$0.a_{31} a_{32} a_{33} \dots$

$\dots$

$\dots$

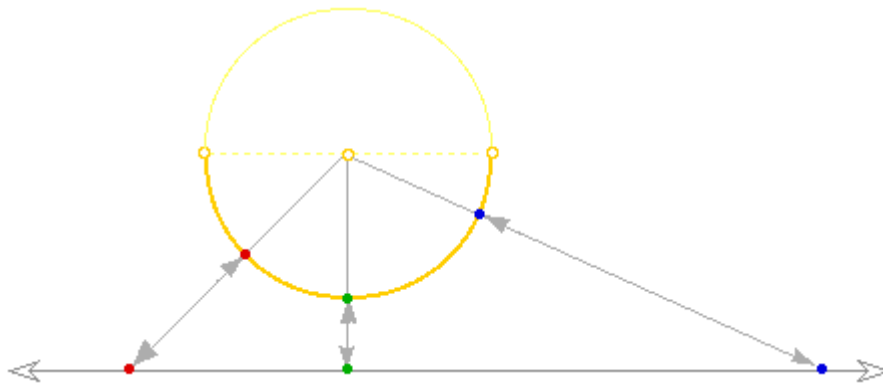
If the assumption that the set of the real numbers in the interval $(0,1)$ is countable, every real number in the interval $(0, 1)$ should be somewhere in this list. Here is how Cantor constructed his great counterexample. The first decimal place in the first number of the above sequence is a_{11} , which is between 0 and 9. Cantor asks us to construct a number as follows. It begins with 0, but the first decimal place would depend on a_{11} – we put any digit different from a_{11} . On the second decimal place we put any digit different from a_{22} , and so on – for every n we construct the n -th decimal place of our number to be *different* from a_{nn} . Notice that the number we constructed according to this *diagonal method* cannot be found on our list – it could not be on the first term of our sequence, since it differs from it at least in its first decimal place; it could not be the second term because at least its second decimal place is different; nor the third, nor the fourth, nor the n -th for similar reasons. In fact, it cannot be anywhere on this list which was assumed to contain all the reals in $(0, 1)$ because it differs from each term of the above sequence in at least one decimal place. Thus our assumption that the set of the real numbers in the interval $(0,1)$ is countable lead to a contradiction!

This simple and elegant proof open the gates to what Hilbert called the *Cantor Paradise*. A set whose cardinality was larger than that of the natural numbers was found! In other words – there exist infinities of different size!

By such striking insights Cantor faced the challenge of authorities like Aristotle, Gauss and Kronecker who said that it was illegitimate to deal with completed (not potential) infinities. By making infinity *actual*, Cantor redefined the notion of “number” in terms of sets and so laid the foundations of modern mathematics.

Questions:

1. Using the figure below can you argue that there are as many points on a short semicircular arc as there are on an entire unbounded line?



2. Can you demonstrate that the infinities of points on two circles of different radius are exactly the same size?

3. Can you demonstrate that the infinities of points on two segments of different length are exactly the same size?

References and further reading

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